Cognitive performance among the elderly in relation to the intake of plant foods. The Hordaland Health Study.

Fruits and vegetables are among the most nutritious and healthy of foods, and are related to the prevention of many chronic diseases. The aim of the study was to examine the relationship between intake of different plant foods and cognitive performance in elderly individuals in a cross-sectional study. Two thousand and thirty-one elderly subjects (aged 70-74 years; 55% women) recruited from the general population in Western Norway underwent extensive cognitive testing and completed a comprehensive FFQ. The cognitive test battery covered several domains (Kendrick Object Learning Test, Trail Making Test--part A, modified versions of the Digit Symbol Test, Block Design, Mini-Mental State Examination and Controlled Oral Word Association Test). A validated and self-reported FFQ was used to assess habitual food intake. Subjects with intakes of >10th percentile of fruits, vegetables, grain products and mushrooms performed significantly better in cognitive tests than those with very low or no intake. The associations were strongest between cognition and the combined intake of fruits and vegetables, with a marked dose-dependent relationship up to about 500 g/d. The dose-related increase of intakes of grain products and potatoes reached a plateau at about 100-150 g/d, levelling off or decreasing thereafter, whereas the associations were linear for mushrooms. For individual plant foods, the positive cognitive associations of carrots, cruciferous vegetables, citrus fruits and high-fibre bread were most pronounced. The only negative cognitive association was with increased intake of white bread. In the elderly, a diet rich in plant foods is associated with better performance in several cognitive abilities in a dose-dependent manner.